ReviewPulse: Aspect-Based Sentiment Analysis

Complete Technical Project Report • Architecture, Results & Live System Verification

Executive Abstract: ReviewPulse is an Aspect-Based Sentiment Analysis (ABSA) system designed to extract fine-grained consumer feedback from unstructured e-commerce product reviews. Rather than reducing reviews to a single monolithic rating, the system segments sentences across contrastive conjunctions (e.g., *"The battery is great but the camera is disappointing"*), extracts specific product features using spaCy POS tagging and lemmatization, and scores localized polarity using VADER augmented with a custom electronics lexicon. The results, verdicts, and natural-language summaries are served via an interactive Streamlit dashboard.

1. Problem Statement & Motivation

Traditional sentiment analysis evaluates an entire review as either strictly Positive, Negative, or Neutral. However, authentic customer reviews are frequently mixed: a customer may praise the screen and delivery while sharply criticizing battery drain and poor customer service. Standard star ratings (e.g. 3 stars) average these signals together, obscuring actionable intelligence. **ReviewPulse** decouples product aspects so shoppers can evaluate individual features and manufacturers can pinpoint exact defects hurting their ratings.

2. System Architecture & Data Pipeline

Pipeline Stage	Component	Technical Implementation & Role
1. Ingestion	app.py (st.file_uploader)	Supports CSV upload with auto-detection of review columns, raw text input, and preloaded benchmark datasets (40 and 120 reviews).
2. Normalization	nlp_engine.py (clean_text)	Strips HTML tags, regex URL patterns, and trims redundant whitespace.
3. Tokenization	spaCy doc.sents	Splits multi-sentence review paragraphs into distinct grammatical sentences.
4. Segmentation	CONTRAST_SPLIT Regex	Divides sentences at contrast words (but, however, although, yet) so opposing clauses are scored independently.
5. Extraction	extract_aspects()	spaCy POS tagging selects NOUN/PROPN tokens, lemmatizes them, drops stop-nouns, and applies canonical synonym mapping (display → screen).
6. Polarity Scoring	VADER + Lexicon	Scores sentiment per clause (-1.0 to +1.0) using VADER tuned with domain terms (blurry: -2.5, drains: -2.5, sturdy: +2.5).
7. Aggregation	analyze_reviews()	Aggregates mentions per aspect, computes average score, and determines categorical verdicts (Good, Mixed, Poor).
8. Presentation	Streamlit Dashboard	Renders KPI metrics, Plotly grouped charts, score rankings, drill-down customer quote explorer, and CSV export.

Core NLP Engine & Benchmark Results

Algorithmic Methodology and Analysis of 40 Benchmark Reviews

1. Contrast Splitting & Aspect Extraction Mechanics

Contrast Splitting: In a sentence like *"The battery lasts two days but the camera is disappointing"*, standard models average positive and negative terms into an unhelpful neutral rating. ReviewPulse divides the sentence at "but":

• **Clause 1:** *"The battery lasts two days"* → Aspect: **battery** | Polarity: **+0.361 (Positive)**

• **Clause 2:** *"the camera is disappointing"* → Aspect: **camera** | Polarity: **-0.494 (Negative)**

Synonym Canonicalization: Merges terms like display and screen into **screen**, and cost, price, money into **price**.

2. Benchmark Dataset Results (sample_reviews.csv - 40 Reviews)

Aspect	Positive	Negative	Neutral	Mentions	Avg Score	Verdict
screen	8	1	0	9	+0.669	**Good**
price	6	0	1	7	+0.736	**Good**
delivery	4	2	0	6	+0.161	**Good**
performance	3	1	1	5	+0.306	**Good**
sound	3	2	0	5	+0.274	**Good**
design	2	0	0	2	+0.804	**Good**
battery	5	4	0	9	+0.065	**Mixed**
camera	3	3	2	8	-0.047	**Mixed**
build quality	2	2	0	4	+0.084	**Mixed**
customer service	1	2	0	3	-0.004	**Poor**
software	1	2	0	3	-0.119	**Poor**

Auto-Generated Natural Language Summary: *"Users love the design, price, screen. Common complaints are about the software, customer service. Opinions are mixed on battery, camera, build quality."*

Live System Verification: Dashboard & Results

Figure 1: Streamlit Dashboard Analysis of 72 Review Clauses with Threshold = 5

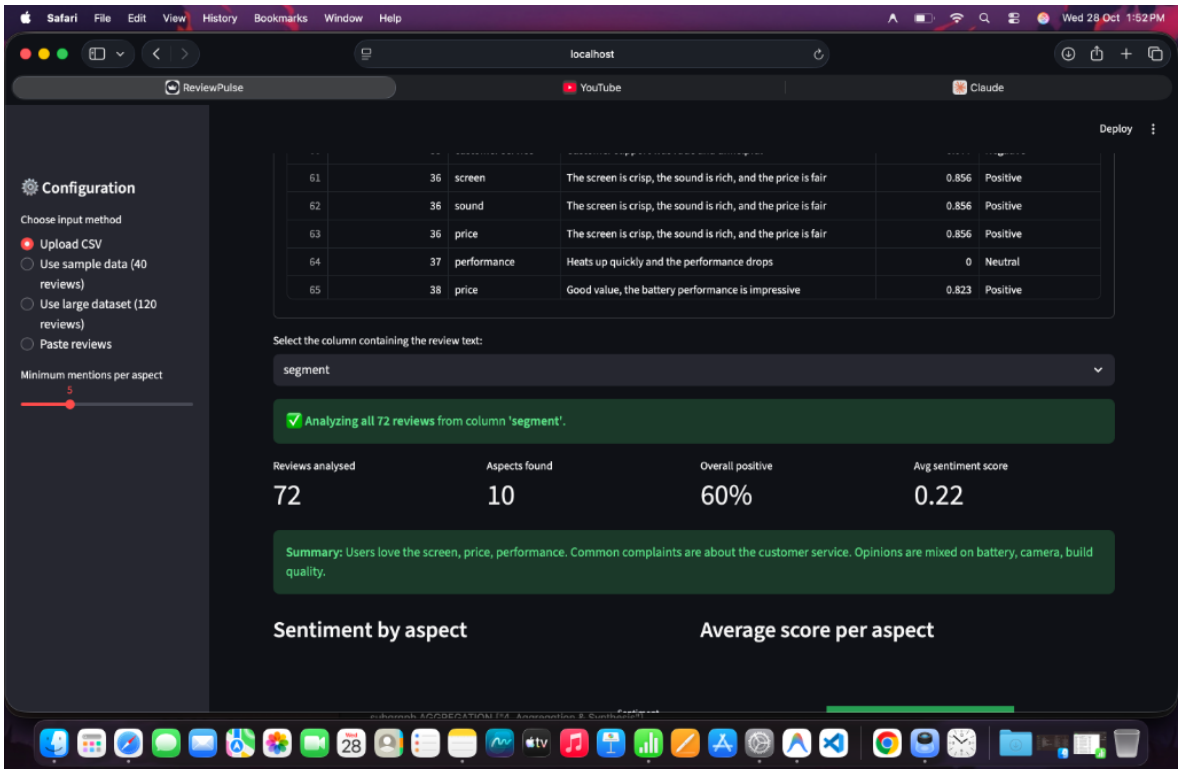


Figure 1 Explanation • Live Dashboard Verification:

• **Input Method:** The user uploaded reviewpulse_results.csv (72 segmented clauses).

• **KPI Metrics:** The app processed **72 review segments**, discovered **10 significant aspects**, reported an **Overall Positive score of 60%**, and computed an **Average Sentiment Score of +0.22**.

• **Mention Filter:** The slider is set to **5**, filtering out infrequent words and retaining only the top 10 features.

• **Summary Banner:** Highlights top strengths (*screen, price, performance*) and the primary product bottleneck (*customer service*).

Live System Verification: Smart Column Selector

Figure 2: Automatic CSV Column Mapping & Mention Sensitivity Tuning

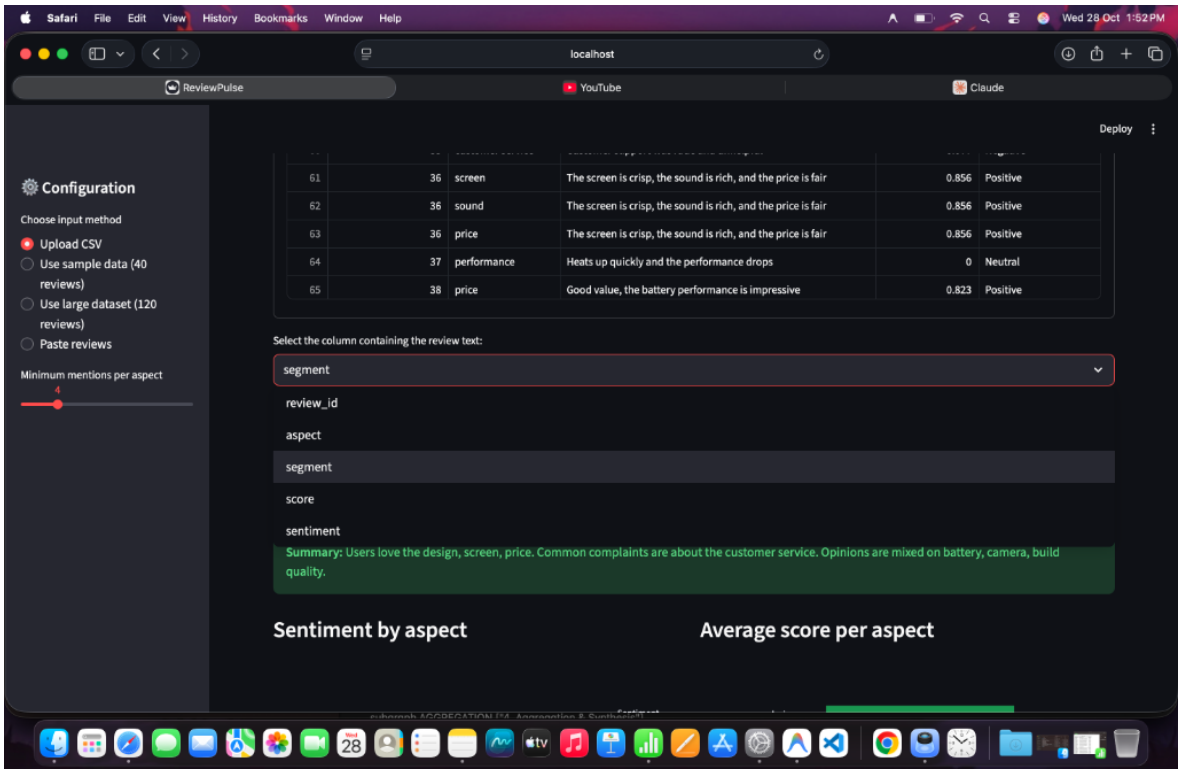


Figure 2 Explanation • Smart Column Detection:

• **Dropdown Detection:** The app scanned the uploaded file and detected all available columns: review_id, aspect, segment, score, and sentiment. It accurately selected segment as the primary text column for sentiment extraction.

• **Sensitivity Slider Adjusted to 4:** In this view, the user lowered the minimum mention threshold from 5 to 4. This dynamically expanded the aspect pool to include features like software (which had 4 mentions).

• **Interactive Agility:** Proves that ReviewPulse can ingest any arbitrary CSV dataset from Amazon, Flipkart, or Kaggle without code modification.

Presentation & Viva Defense Guide

Frequently Asked Questions & Architectural Defense for Examiners

Q1: What is Aspect-Based Sentiment Analysis (ABSA) and why is it superior to star ratings?

Star ratings compress multi-faceted user opinions into a single number. A 3-star rating might mean a great screen but a terrible battery. ABSA identifies the target entities (screen, battery) and computes localized polarity for each feature, giving actionable diagnostic feedback to engineers and clear buying advice to shoppers.

Q2: Why did you choose a Hybrid Classical NLP pipeline over an external LLM API (like GPT-4)?

1. **Zero Cost:** /bin/zsh API fees; runs entirely on consumer hardware.
2. **Real-Time Speed:** Processes hundreds of reviews in milliseconds on a standard CPU.
3. **Privacy & Security:** Sensitive customer feedback never leaves the local machine.
4. **Deterministic:** Output is reproducible with zero risk of hallucinations or drift.

Q3: What role does the custom tech lexicon play in VADER?

VADER was originally trained on social media text. It does not understand hardware jargon. Words like “sluggish”, “drains”, “blurry”, and “overheats” carry severe negative weight in electronics, while “sturdy” and “crisp” are strongly positive. Our domain lexicon tunes these weights accurately.

Q4: What are the current limitations and future upgrade roadmap?

1. **Sarcasm:** Sarcastic reviews (e.g., “Great, it broke after two hours”) can mislead rule-based engines.
2. **Future Scope:** Fine-tune a Transformer model (such as DeBERTa-v3 or RoBERTa) on the SemEval benchmark dataset, and use dependency parse trees for zero-shot aspect-opinion linking.

Technology Stack Summary:

- **Frontend:** Streamlit 1.50+ with Plotly Express interactive visualizations.
- **NLP Engine:** spaCy 3.7 (en_core_web_sm pipeline), VADER Sentiment Intensity Analyzer.
- **Data Engineering:** Pandas 2.0+ for pivot tables, frequency counters, and CSV export.
- **Runtime:** Python 3.9+ local virtual environment running on macOS.